Phase 1: The Input Data

- **What you need:** A free dataset of user comments, product reviews, or forum posts.
- **Where to get it:** Download a public CSV file from Kaggle (such as YouTube comments, Reddit threads, or product feedback datasets).
- **What the raw data looks like:** A spreadsheet with at least two columns:
    - text_id (A unique number for each comment).
    - raw_text (The actual comment or review written by a user).

Phase 2: Data Cleaning (Preprocessing)

Raw text is messy. Clean it before analysis using Python.

- **Lowercase conversion:** Change all letters to lowercase so they are treated the same.
- **Noise removal:** Strip out URLs, emojis, HTML tags, and extra spaces.
- **Tokenization:** Split each sentence into individual words.
- **Stop-word removal:** Delete filler words that carry no emotional weight (like "the", "is", "at").

Phase 3: The 3-Tier Classification Pipeline

This is the core logic. Process texts through three strict stages.

Tier 1: Broad Intent Sorting

Every cleaned text is sorted into one of three clean buckets:

- **Actionable / Constructive:** The user wants to improve something (suggestions, ideas, direct feedback).
- **Informational:** Neutral statements, personal stories, general observations, and pitfalls.
- **Hostile / Toxic:** Sarcasm, polite insults without cussing, direct attacks.

Tier 2: The Innovation Layer (Sarcasm & Veiled Insults)

Take only the texts that fell into the Hostile / Toxic bucket from Tier 1 and run them through a specialized check:

- **Sarcasm Detection:** Look for mismatched words. (Example: Using positive words like "Oh, brilliant" combined with a negative context).
- **Veiled Insult Detection:** Look for polite phrasing carrying hostile weight.
- **Output:** Label them specifically as Sarcastic or Non-Profane Insult instead of generic "negative".

Tier 3: Value Extraction

You keep the category structure simple by using a flexible extraction rule:

- **Default Extraction:** The Actionable / Constructive bucket goes under value extraction by default.
- **User-Choice Extraction:** The Informational category (including personal stories and pitfalls) and other categories can be pulled into value extraction based on the user's explicit choice.
- **Inside Extraction:** Filter out generic praise ("Great video!"). Keep actual ideas, suggestions, experiences, and troubleshooting lessons so a developer or creator knows what to fix or build.

Phase 4: Big Data Integration (For the Syllabus)

Tie this to Big Data technologies:

- **Storage:** Store your raw CSV dataset and final processed results inside HDFS.
- **Processing:** Use Apache Spark locally via Python (PySpark) to handle and process rows in parallel.
- **Querying:** Use Hive to run SQL-like summary queries on your final categorized results (e.g., counting how many sarcastic comments appeared per day).

Phase 5: Documentation & Presentation

- **Report Writing:** Document your 3-tier pipeline, showing why traditional sentiment analysis lacks and how the innovation improves it.
- **The Video Requirement:** Record a short 2 to 5-minute video demonstrating your code running and showing the final categorized output.